

**Think Before You Share:
A Technology-Enabled Approach to
Combating Vaccine Misinformation
on Social Media**

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I. Introduction

Social media has become an important source of information for many people, particularly on health and vaccination. It allows health information to reach large audiences quickly and makes communication more accessible. However, the same platforms can also contribute to the rapid spread of false, misleading, or incomplete information. Vaccine misinformation can make it difficult for individuals to distinguish reliable health information from unsupported claims.

The World Health Organization (WHO) describes an excessive amount of information, including false or misleading information, as an “infodemic.” This can create confusion, encourage harmful health decisions, reduce trust in health authorities, and weaken public-health responses (World Health Organization, n.d.). Regarding vaccination, misinformation may increase fear and uncertainty, contributing to vaccine hesitancy. A review by Skafle et al. (2022) found that vaccine misinformation on social media commonly involved medical misinformation, vaccine development, and conspiracy theories. The reviewed studies generally indicated that exposure to such misinformation was negatively associated with vaccine acceptance and uptake.

This concern is also relevant in the Philippines. UNICEF has recognized that misinformation circulating through social media can contribute to fears and misconceptions about vaccines. Efforts to improve confidence in routine immunization have therefore included using social media and community-based communication to provide reliable information to the public (UNICEF, n.d.). Given the widespread use of digital platforms, technology can also be used to make credible vaccine information more accessible and to encourage individuals to verify health information before sharing it.

II. Proposed Solution

To address vaccine misinformation on social media, this project proposes three technology-enabled approaches: a Vaccine Misinformation Reporting System, a “Verify Before You Share” Browser Tool, and a QR-Based Vaccine Fact Check. These solutions are intended to

improve access to reliable vaccine information and encourage users to evaluate the credibility of health-related content before sharing it online.

The Vaccine Misinformation Reporting System would provide an online platform where users can submit suspicious vaccine-related posts, claims, links, or screenshots for review. The submitted information would be checked against credible sources from recognized health authorities such as the Department of Health, World Health Organization, and UNICEF. The platform could provide clear explanations regarding questionable information and direct users to reliable sources where they can learn more about the topic.

The “Verify Before You Share” Browser Tool would serve as a digital reminder for users who encounter vaccine-related information online. The browser extension or mobile tool would encourage users to assess a source's credibility before sharing vaccine-related content. It could also provide convenient access to reliable health websites, allowing users to compare information with evidence-based resources. This approach is intended to promote responsible social media use and reduce the unintentional sharing of misleading information.

The QR-Based Vaccine Fact Check would provide another simple, accessible way to obtain reliable vaccine information. QR codes could be displayed in health centers, schools, vaccination sites, pharmacies, hospitals, and public information materials. After scanning the QR code using a smartphone, users would be directed to a mobile-friendly webpage containing verified vaccine information, frequently asked questions, explanations of common misconceptions, and links to trusted health authorities. The content would need to be regularly reviewed and updated to ensure that the information remains accurate and relevant.

Among the proposed approaches, the QR-based system offers a practical option because QR codes are inexpensive to produce and can be accessed through most smartphones without requiring users to install a separate application. When combined with the reporting system and browser tool, these technologies can provide several opportunities for individuals to verify vaccine information and develop more responsible information-sharing practices.

III. Conclusion

Vaccine misinformation on social media is a significant public-health concern because misleading information can contribute to fear, confusion, vaccine hesitancy, and reduced trust in vaccines and healthcare institutions. The rapid circulation of information through digital platforms makes it increasingly important for individuals to have convenient access to reliable and evidence-based health information.

The proposed Vaccine Misinformation Reporting System, “Verify Before You Share” Browser Tool, and QR-Based Vaccine Fact Check demonstrate how technology can be used to support the responsible use of vaccine information. These solutions can provide users with easier access to credible sources, encourage verification before sharing, and improve awareness of the importance of evaluating online health information.

The effectiveness of these solutions will depend on the accuracy and regular updating of their content, the appropriate management of submitted information, and coordination with credible health authorities. Technology should not replace professional medical advice but should serve as an additional tool for improving access to trustworthy information. Through responsible implementation and continued public education, these proposed solutions can help reduce the influence of vaccine misinformation and promote informed health decisions.

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